

**Evaluating the Contribution of Road Network Topology to Surface Urban Heat Island
Estimation in Davao City Using Graph Convolutional Networks**



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CHAPTER I

INTRODUCTION

Urbanization and changes in land use have intensified environmental challenges in cities facing the combined pressures of development and climate change. As urban populations grow, more residents are exposed to elevated temperatures, increasing the need for locally appropriate heat-mitigation measures. A global assessment of future urban temperatures emphasizes the potential role of nature-based solutions in limiting these impacts [1]. Satellite evidence also shows that urban surface warming can exceed warming in surrounding areas: an analysis of more than 2,000 city clusters found that the mean surface warming trend in urban cores was 29% greater than in rural backgrounds during 2002-2021 [2]. The combined influence of urban development and climate change has been investigated through complementary methods. A study of Lagos separated urbanization-related and broader climatic contributions to changes in urban surface temperature, demonstrating the importance of distinguishing local land-cover effects from background warming [3]. Scenario-based research in Tehran also examined urbanization and climate change in relation to urban heat and thermal comfort [4].

At the global scale, projections integrating future urban expansion and climate change indicate that urbanization can add to local surface warming through 2100 [5]. Land surface temperature (LST) describes the thermal condition of the ground, vegetation, roofs, and other surfaces as estimated from thermal infrared observations [6]. It differs from the temperature of the overlying air. A surface urban heat island (SUHI) occurs when urban surfaces are warmer than a specified surrounding non-urban background; SUHI intensity expresses this difference rather than the absolute surface temperature [7]. Consequently, high LST alone does not establish the magnitude of SUHI intensity. The reference surface must be defined explicitly because differences in rural land cover can change the resulting urban-rural temperature contrast [8]. Research comparing crowdsourced air temperatures with satellite observations further shows that surface and air-temperature heat islands can differ in magnitude and their relationships with environmental characteristics [9]. The relevance of urban thermal information extends to adaptation planning. A review of resilience to weather and climate extremes identifies infrastructure, socioeconomic conditions, and governance as interacting influences on urban adaptive capacity [10]. Research on urban heat-risk management similarly emphasizes differentiated vulnerability, participation, uncertainty, and

the sequencing of adaptation actions [11]. Consequently, identifying warmer urban surfaces is a useful component of environmental assessment, but deciding where and how to intervene requires evidence about local conditions and the people affected. This distinction helps define the contribution of a surface-temperature model within a broader planning process.

Southeast Asia provides an important regional setting for urban thermal research. A regional analysis examined thermal discomfort days from 1940 to 2020 and identified increasing exposure across the region, with different patterns between continental and maritime countries [12]. This analysis concerns thermal-comfort indices rather than SUHI intensity, but it provides a regional context for the need to investigate urban thermal conditions. At the surface level, a comparative study examined urbanization and climate-related thermal variation in Bangkok and Ho Chi Minh City [13]. A more recent multi-decadal study of Southeast Asian capitals also documented land-cover changes alongside surface-temperature trends [14]. Furthermore, regional projections reinforce the need for this distinction. Climate-model analyses indicate that population exposure to heat extremes in Southeast Asia increases at higher global warming levels, with exposure affected by both climatic and demographic change [15]. A separate assessment combines the Universal Thermal Climate Index and Wet-Bulb Globe Temperature with population distributions to examine future heat stress across the region [16]. These indices incorporate information relevant to human thermal stress that cannot be recovered from LST alone. They establish the broader urgency of heat adaptation without supplying a direct substitute for the surface-temperature target used in this study. Surface-temperature patterns also vary within and between tropical cities. An analysis of 19 Southeast Asian cities along urban-rural continuums relates thermal changes to built-up expansion and vegetation change, while identifying differences across levels of urbanization [17]. Research comparing an inland and a coastal Thai city reports different spatial and daytime-nighttime thermal patterns [18]. Beyond Southeast Asia, a MODIS-based analysis of South Asian ecoregions documents geographically and seasonally varying warming and cooling trends [19].

Furthermore, the regional literature also identifies limits in translating temperature assessments into adaptation. A review found that Southeast Asian urban heat research concentrates heavily on satellite and meteorological characterization, while the socioeconomic dimensions affecting vulnerable communities receive less attention [20]. Research on heat-adaptation financing further identifies financial constraints that shape the implementation of heat-adaptation measures [21]. Moreover, evidence on mitigation further

connects spatial assessment with practical planning. A review of Southeast Asian heat-island mitigation examines green infrastructure and other cooling strategies alongside their implementation feasibility [22]. Research combining satellite analysis and participatory mapping in Metro Manila and Bangkok also shows how community knowledge can identify locally experienced hotspots and exposure pathways that coarse thermal observations miss [23]. Together, these approaches suggest that mapped surface conditions can help prioritize more detailed assessment, while implementation and evaluation should incorporate field observations and local experience. The present study contributes to the mapping stage and does not test the effectiveness of a cooling intervention.

In the Philippines, long-term meteorological records provide evidence of changes in temperature extremes. A national assessment analyzed daily precipitation and near-surface air temperature for 1951-2010 [24]. A subsequent analysis of observations from 34 meteorological stations for 1951-2018 identified differences in temperature trends between urban and rural stations, with stronger warming in indices associated with minimum temperature [25]. These atmospheric observations establish a national context, but they should not be treated as direct measurements of satellite-derived SUHI intensity or as proof that every observed temperature change is caused by urbanization. Philippine remote-sensing studies provide more direct evidence of surface thermal variation. An analysis of Metropolitan Manila found that increasing built-up indicators and declining vegetation indicators were associated with higher LST in Metropolitan Manila between 2001 and 2019 [26]. A study of Manila City combined satellite and meteorological information to assess intra-urban thermal variation in Manila City and locate areas for further intervention assessment [27]. Beyond the capital region, research examined land-cover changes in the Cagayan de Oro River basin from 2003 to 2020 and reported higher mean LST over built-up areas and bare land than over forest, mangroves, and water bodies [28]. In fact, a localized heat-risk assessment in Quezon City integrates environmental exposure and socioeconomic vulnerability in a weighted index, illustrating how densely developed and vegetation-poor areas can overlap with socially vulnerable communities [29].

Atmospheric modeling offers complementary evidence about the Philippine urban environment. A numerical simulation study used the Weather Research and Forecasting model coupled with an urban canopy model to examine urban land cover and anthropogenic heat during extreme heat events in Metro Manila [30]. A related study evaluated urban canopy model configurations for air temperature and relative humidity during a high-heat

event, emphasizing the importance of appropriate urban morphology information [31]. These studies concern atmospheric processes and do not establish that heat travels along road-network connections. At the national scale, a heat-health assessment combined MODIS-derived thermal information with population data for a heat-health assessment [32].

These thermal pressures are also evident in Davao City. A local study mapped the occurrence of hot and cold surface-temperature locations from 1994 to 2019 and examined associations with vegetation, built-up indicators, albedo, elevation, and population density. Its quantile-regression results identified the Normalized Difference Built-up Index (NDBI) as the most significant predictor at the 50th, 75th, and 90th LST quantiles, while other predictors were more important at lower quantiles [33]. This variation cautions against assuming that a single environmental factor explains thermal conditions throughout the city. Another local investigation reported higher surface temperatures in densely built-up, vegetation-deficient parts of Davao than in greener areas [34]. These local studies establish a basis for further surface-temperature modeling, but they do not determine whether road-network connectivity adds predictive information beyond the environmental attributes observed at each location. Additional Landsat-based research in Davao City for 2018-2022 reports a positive association between NDBI and LST and a negative association between NDVI and LST [35].

Satellite remote sensing provides spatially extensive information for examining these local differences, subject to cloud cover, retrieval uncertainty, and the spatial and temporal characteristics of the observations [6], [7]. Machine-learning approaches used in urban heat research include Random Forest (RF), convolutional neural networks (CNNs), and other neural architectures [36]. A further comprehensive review examines the range of machine-learning methods used for urban heat-island assessment [37]. However, predictive performance depends on how the data, spatial relationships, and evaluation procedure are represented. A review of environmental spatial machine learning identifies several ways to account for spatial autocorrelation, including spatial covariates and training-test partitioning, and finds no single approach that is appropriate for every application [38].

Empirical comparisons provide a reason to retain a conventional benchmark. Fine-scale urban heat modeling has compared Random Forest with multiple linear regression using air temperature and a broad set of environmental and urban predictors [39]. Although the response variable differs from satellite LST, the study demonstrates the relevance of evaluating nonlinear models against alternatives under a defined task. An open-source

UHI-mapping framework for Lagos uses a U-Net architecture with satellite and urban geospatial inputs [40], illustrating a raster-based approach to a related thermal problem.

Although Random Forest can capture nonlinear relationships between environmental predictors and LST, a conventional tabular implementation does not explicitly aggregate information through road-network connections unless such information is encoded in its inputs. Standard raster CNNs represent local spatial context through grid neighborhoods; those neighborhoods do not explicitly distinguish locations connected by a road from locations that are merely close in geographic space. A road graph provides a different representation: intersections and intermediate road points become nodes, and connecting road segments define their adjacency. Street-graph research demonstrates the usefulness of network-based units for characterizing urban morphology, although that work uses geometric information beyond connectivity alone [41]. In this study, road adjacency is treated as a candidate relational representation whose predictive contribution must be tested. It is not assumed to be a physical heat-transfer pathway or a complete description of street-canyon geometry.

A graph representation also requires a clear interpretation of its edges. Research on regional surface heat islands has combined morphological spatial pattern analysis and graph-based connectivity measures to evaluate the configuration of thermal patches [42]. Such a thermal-landscape graph differs from a street graph because its connections are defined for a different analytical purpose. This distinction matters when adapting graph methods: an adjacency matrix is a modeling choice, and evidence obtained with thermal patches cannot by itself validate road adjacency. In the proposed model, connected road locations provide neighborhoods for information aggregation, with their usefulness assessed through prediction errors.

Graph neural networks (GNNs) offer mechanisms for learning from both node attributes and the relationships specified by a graph [43]. Environmental applications provide relevant methodological precedents. One study developed graph-based spatiotemporal interpolation using a Voronoi-adjacency representation [44]. Another study applied a GCN framework to daytime and nighttime SUHI modeling in Hong Kong, integrating built and natural urban features [45]. A recent review examined graph construction and spatial-temporal learning across several urban environmental applications [46]. These studies support investigating graph-based thermal estimation, but results obtained with other graph

definitions, environmental variables, and locations cannot establish the benefit of Davao City's road topology in advance.

Spatiotemporal graph methods offer several ways to combine relationships between locations with changes through time. Spatial-Temporal Synchronous Graph Convolutional Networks model localized spatial and temporal relationships in network data [47]. Work on scalable spatiotemporal autoregressive modeling also connects spatial-dependence models with graph convolutional neural networks [48]. A systematic review of spatiotemporal GNNs identifies comparability, reproducibility, explainability, and scalability among the challenges facing these approaches [49]. More directly related thermal applications include graph-neural-network forecasting of road surface temperature [50]. Environmental applications also include a variable-resolution spatiotemporal GNN that generates very short-range forecasts of near-surface atmospheric variables in Guangdong [51]. Their relevance is methodological: graph models can be formulated for geographically distributed temperature-related observations. Road-surface forecasting and hourly atmospheric forecasting nevertheless differ from estimating annual satellite-derived LST. Their results, therefore, motivate investigation without justifying claims about short-term forecasting, subpixel road temperatures, or physical heat movement in the present study.

The present study will represent road-associated locations in Davao City using an OpenStreetMap (OSM) network acquired and processed with OSMnx, which provides methods for constructing and analyzing street graphs [52]. Nodes will include intersections and intermediate points along road segments. Each node will be associated with historical LST, the Normalized Difference Vegetation Index (NDVI), and NDBI. NDVI and NDBI will be treated as spectral indicators of vegetation and built-up characteristics, respectively; NDBI will not be interpreted as a direct measurement of impervious-surface density. Road connectivity will allow the model to combine information from connected locations. The hypothesis is that this relational information may contribute to estimation after accounting for the historical environmental features at each node. Combining OSM with satellite and environmental information has also been explored in a multi-domain urban-planning framework that addresses land use, building footprints, traffic, and air quality [53]. This provides a precedent for integrating volunteered geographic information with remotely sensed inputs, while the present study has a narrower thermal-estimation purpose. OSM coverage and the alignment of road geometry with satellite observations will consequently be

documented as properties of the input data. A more detailed road graph will not be treated as evidence of equally detailed thermal measurements.

The proposed architecture combines a Long Short-Term Memory (LSTM) encoder with GraphSAGE neighborhood-aggregation layers. LSTM provides the temporal sequence-learning component [54], while GraphSAGE provides a method for learning node representations by aggregating neighborhood information [55]. In this chapter, the specific model is called LSTM-GraphSAGE to distinguish its architecture from the broader GNN family. Annual median environmental features from 2023 and 2024 will be used to estimate the 2025 LST associated with each node. SUHI intensity will then be derived for urban nodes by subtracting a common, independently specified non-urban reference LST for 2025. This historical target-year experiment evaluates spatial estimation using prior-year features; it does not establish operational short-term forecasting or reliable prediction under future infrastructure changes.

The methodological gap is whether explicitly representing road-network connectivity improves estimation beyond what can be learned from the same historical satellite-derived features without those connections. Comparing LSTM-GraphSAGE with Random Forest alone would not isolate this contribution because both the architecture and the use of adjacency would change. The study will therefore compare the proposed road-connected configuration with a matched configuration in which inter-node road connections are disabled, while retaining the node's own information and the same temporal encoder, layer dimensions, and training procedure. A Random Forest model using the same historical predictors will provide a separate conventional benchmark. All configurations will be assessed on the same held-out observations.

Through these comparisons, the study will evaluate both overall estimation performance and the added predictive value of the chosen road-network representation. Its outputs will include estimated LST, derived SUHI intensity at urban road-network nodes, and maps of locations with elevated estimated SUHI intensity. These outputs are intended to support the interpretation of surface thermal variation within the defined urban study area of Davao City. Neither improved performance nor the ability to identify hotspots will be assumed before evaluation.

Research Questions

This study seeks to evaluate whether road-network topology contributes useful information to the estimation and mapping of satellite-derived SUHI intensity in Davao City. Specifically, it will address the following questions:

1. How accurately can the proposed LSTM-GraphSAGE model, integrating OSM road-network topology and historical satellite-derived environmental features, estimate LST and the resulting SUHI intensity at urban road-network nodes in Davao City?
2. What change in estimation performance results from incorporating road-network connectivity compared with a matched model configuration in which inter-node road connections are disabled?
3. How does the predictive performance of the proposed model compare with a Random Forest regression baseline using the same historical predictors and held-out observations, based on root mean square error (RMSE), mean absolute error (MAE), and the coefficient of determination (R^2)?
4. How effectively can the proposed model identify and spatially locate areas of elevated SUHI intensity at urban road-network nodes, relative to the corresponding held-out satellite-derived observations?

Significance of the Study

This study will provide spatially explicit information on SUHI intensity and an empirical assessment of road-network topology as a representation for thermal estimation. Its potential contributions concern urban planning, local environmental assessment, and reproducible geospatial machine learning.

GeoAI and Computer Science. The comparison between road-connected and connectivity-disabled configurations will examine the predictive value of an explicit network representation while controlling the principal architectural differences. The separate Random Forest benchmark will assess the complete model against a conventional machine-learning approach. A finding that road connectivity provides little or no improvement would also be informative because it would clarify the limits of this representation for the selected data and task.

Remote Sensing and Urban Climate Research. The study will relate satellite-derived thermal and spectral information to road-associated locations while maintaining the distinction between LST and reference-based SUHI intensity. Documenting the reference-area definition, imagery coverage, and spatial support of the estimates will help make the results interpretable and comparable with subsequent local studies.

Urban Planning and Engineering. The resulting maps may help the City Planning and Development Office and the City Engineer's Office identify road-associated locations with elevated estimated SUHI intensity for further site assessment. Such information may inform the consideration of shading, vegetation, and surface-material interventions alongside field observations and engineering requirements. The study will not determine the optimal intervention or demonstrate the cooling effectiveness of cool pavements, permeable materials, or urban greening.

Local Government and Environmental Management. The study may provide supporting spatial evidence for discussions of land-use planning, zoning, and urban greening priorities. The maps can indicate where further environmental investigation is warranted, but they will not constitute a heat-health warning system. Decisions involving disaster response or public health would require additional meteorological, exposure, and vulnerability information.

Open and Reproducible Geospatial Modeling. The use of OSM, Landsat data, and documented processing procedures will support reproducibility. Recording acquisition dates, quality-screening rules, graph-construction decisions, model settings, data partitions, and evaluation procedures will enable other researchers to examine the workflow and assess its suitability for comparable urban contexts.

Scope and Delimitations

The geographic extent of road-network acquisition will follow the Davao City administrative boundary obtained from OSM. This acquisition extent includes urban, agricultural, rural, and mountainous areas; however, citywide acquisition does not mean that every road node represents an urban SUHI observation. The principal estimation and interpretation will focus on nodes within the continuously built-up urban area. Urban evaluation locations and non-urban reference surfaces will be distinguished through documented geographic and land-cover masks. Areas will not be designated urban, excluded, or retained solely to obtain a more connected graph. The exact masks, road-type filters, and

treatment of disconnected components will be specified in the methodology and reported with the resulting coverage.

The study will use Landsat 8 Collection 2 Level-2 surface reflectance and surface temperature products for 2023-2025. The USGS surface temperature product provides the thermal observations, while NDVI and NDBI will be computed from the corresponding reflectance data using the appropriate product scaling and quality information [56]. Cloud, cloud-shadow, and invalid observations will be screened before constructing annual median composites. Historical LST, NDVI, and NDBI from 2023 and 2024 will form the model inputs, and the 2025 LST composite will provide the target observations. Valid-scene counts, valid-pixel coverage, and differences in observation availability will be documented for each year. The analysis is restricted to this Landsat 8 workflow; multisensor fusion with Landsat 9 or ECOSTRESS is outside the defined comparison.

The model will estimate LST, from which urban-node SUHI intensity will be derived. For an urban node i and target year t , the operational relationship is $SUHI(i,t) = LST(i,t) - LST_{ref}(t)$, where $LST_{ref}(t)$ is the median of valid annual-composite LST pixels within the designated non-urban reference mask for the same year. Estimated SUHI will use the model-estimated urban LST and the same reference value used for the satellite-derived comparison. The reference mask will be defined independently of model prediction errors and will retain non-urban reference pixels even when they are outside the modeled urban graph. Its land-cover composition and elevation comparability with the urban area will be documented, and water surfaces will not be pooled into the terrestrial rural reference. Because reference selection affects SUHI interpretation, the reference boundary and eligibility criteria must be fixed before final evaluation [8]. This procedure produces a reference-based surface-temperature contrast, not an independently measured air-temperature heat island.

The road graph will contain intersections and intermediate points along retained road segments. Node densification changes the locations at which features are assigned and predictions are reported; it does not create finer independent thermal observations. Spatial detail remains constrained by the native and processed support of the Landsat observations. Accordingly, references to the street-network level describe the placement of estimates on a road graph, not direct measurement of pavement temperature or pedestrian-level microclimate. The road-network representation will also depend on OSM completeness,

positional accuracy, attribute quality, and the date of extraction. An OSM network snapshot will not be assumed to reproduce all road changes occurring during 2023-2025.

The comparative analysis will evaluate three configurations: LSTM-GraphSAGE with OSM road adjacency, a matched configuration with inter-node road connections disabled, and Random Forest regression. The two neural configurations will share the temporal inputs, layer dimensions, optimization settings, and evaluation observations. Random Forest will receive the same historical information in tabular form. Target-year LST at held-out nodes will not be included among their predictor features or used for model selection. Because nearby nodes may share satellite pixels or spatially correlated observations, the data-partitioning procedure will account for shared observation support and spatial dependence; otherwise, apparent test performance could overstate generalization [38]. The methodology will distinguish any use of test-node historical features in graph computation from access to their held-out target labels.

Performance will be reported using RMSE, MAE, and R^2 on identical held-out observations. Applying one common reference value to both predicted and observed LST leaves the corresponding SUHI prediction errors unchanged for a single target year; LST and SUHI scores derived in this way will therefore not be presented as independent validations. Hotspot assessment will use a classification rule established without tuning to the held-out test results, and predicted hotspot locations will be checked against the same rule applied to held-out satellite-derived SUHI values. A visually plausible map alone will not establish hotspot-detection accuracy.

The temporal scope consists of two historical annual input steps and one target year. It is insufficient to establish long-term temporal generalization, seasonal behavior, diurnal heat retention, or short-term heat-event forecasting. Annual medians summarize the available quality-screened satellite observations rather than continuous all-weather conditions. The study excludes ground-based air-temperature prediction, thermal comfort, heat index, human heat exposure, and health-risk estimation. It will not simulate physical heat transport along roads, quantify the causal effects of individual road connections, or predict the cooling outcomes of proposed infrastructure changes. Its conclusions will be limited to comparative estimation performance and mapped surface thermal conditions under the documented data and reference definitions.

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